MIL-T-82332A
1 March 1968
SUPERSEDING
MIL-T-82332 (ONR/NTDC)
15 July 1965
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13 September 1963

MILITARY SPECIFICATION

TRAINING AID, ENLARGED DEAD RECKONING COMPUTER

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for the production of an Enlarged Dead Reckoning Computer for use as a training aid.

1.1.1 This item has been designated by the Naval Training Device Center as Device 1GG18A (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or requests for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-621	Boxes, Wood, Nailed and Lock-corner

Military

MIL-P-116	Preservation; Methods of
MIL-L-10547	Liners, Case and Sheet, Overwrap; Water-vaporproof or Waterproof, Flexible

FSC-6910

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STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property

DRAWINGS

Naval Training Device Center

Drawing No. 3313-65-404 (sheets 1 through 19)	Device 1GG18A

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Materials.- Unless otherwise specified herein, materials shall be as specified on the drawings of 2.1. Materials not specified herein or on the drawings shall be selected on the basis of compatibility with the established design and suitability to the specific application in satisfying the performance requirements specified herein.

3.1.1 Lacquers or inks shall be those which will form a permanent bond with the material surface.

3.1.2 All metal parts and fastenings shall be of stainless steel, or be satin-finish chromium-plated or anodized clear, as applicable.

3.1.3 Plastic.- Plastics shall be durable and resistant to cracking, brittleness, discoloration and deformation upon aging, or when subjected to the range of temperatures specified herein. Transparent panels shall be made of acrylic plastic.

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3.1.3.1 Cementing.- Cements used shall be those recommended by the manufacturer of the plastic. The cemented joints shall be free of air bubbles and form a strong, homogeneous, permanent bond.

3.1.4 Hardboard.- Hardboard shall be of the tempered variety, smooth on both sides, hard surfaced and moisture resistant for a paint finish.

3.2 Design.- The training aid shall be suitable to demonstrate the use of the operational computer to a group of students in a classroom. The enlarged computer shall be five times the size of the operational computer. One face of the enlarged computer shall consist of compass rose, plotting area and vertical slide. The other face shall be represented by the dead reckoning computer. Both faces shall be mounted side by side on a transverse hardboard backing.

3.2.1 The plotting area shall consist of a clear sheet of acrylic not less than 0.100 inch thick. The rotating disc shall be supported by four nylon rollers. The vertical, airspeed grid slide shall carry markings on both sides. The guides for this slide shall allow the easy removal and reinsertion of the slide. A thumb-screw locking device shall hold the slide in any position.

3.2.2 The computer side shall consist of one stationary and one rotating disc.

3.2.3 All faces of the training aid shall be painted satin-finish white and silkscreened or imprinted in black with the scales and lines of the operational computer in the proper enlargement.

3.3 Construction.- The training aid shall be constructed in accordance with the drawings specified in 2.1 and the following:

3.3.1 Construction methods or materials may be varied in order to permit more practical and economical manufacturing processes and tooling practices. The changes will be acceptable provided that they do not adversely affect the training value or durability of the training aid. All changes shall be submitted in the form of preliminary drawings or data for written approval by the Contracting Officer prior to the construction of the training aid.

3.4 Performance.- The training aid, including all components, shall be constructed to perform in accordance with the requirements of this specification under ambient conditions of temperature ranging from minus 15 degrees centigrade to plus 55 degrees centigrade and of relative humidity up to 95 per cent. It shall also withstand shock and vibration encountered in normal classroom use and, when properly packed, during shipment by common carrier. No fixed part or assembly shall become loose, no movable part or assembly shall become undesirably free or sluggish in operation, and no degradation shall be caused in the mechanical performance of the training aid due to shipment, operation, or storage.

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3.5 Details of components.- The training aid shall consist of the following major components, the details of which shall be in accordance with the drawings of 2.1:

(a) Baseplate

(b) Compass rose, plotting area, and airspeed grid slide

(c) Computer

3.6 Dimensions and tolerances.-

3.6.1 Dimensions.- The overall dimensions shall be approximately 56 by 50 by 2 inches.

3.6.2 Tolerances.- All tolerances shall be such that the training aid operates freely by hand without binding or excessive friction. Computations carried out with the device shall have the same numerical accuracy as obtainable with the operational computer.

3.7 Finish and color.-

3.7.1 All surfaces shall correspond in color and texture to color chip 27875, Federal Standard No. 595.

3.7.2 All grids, scales, lines and lettering shall be in black. Indications may be produced by printing or silkscreening.

3.7.3 All printing shall be covered by a protective transparent lacquer or equivalent transparent film. The coating shall form a permanent bond with the imprint and the imprinted surface.

3.8 Nameplates or product markings.- Unless otherwise specified by the procuring activity, a nameplate prepared in accordance with MIL-STD-130 shall be affixed to the training aid. Additional identification, when required, shall be furnished by the procuring activity. (see 6.2(e)).

3.9 Workmanship.- The equipment, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be paid to neatness and thoroughness of marking of parts and assemblies, plating, sanding, grinding, painting, and riveting.

3.9.1 Fabrication.- Machining, drilling, and forming shall be done with the use of accurate templates, jigs or gages.

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3.9.2 Riveting.- Riveting operations shall be carefully performed in order to insure that the rivets are tight and do not deform the material.

3.9.3 Cleanup.- Units shall be thoroughly cleaned after final assembly. Burrs and sharp edges shall be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspections to be performed are classified as follows:

(a) In-process inspection

(b) Quality conformance inspection.

4.3 In-process inspection.- In-process inspection shall include such visual and mechanical examinations and testing of materials, subassemblies, parts, and accessories (including purchased standard items) during the manufacturing process as may be necessary to assure conformance to all the requirements specified herein.

4.4 Quality conformance inspection.-

4.4.1 Preproduction inspection.- Preproduction inspection shall be conducted on the preproduction sample and shall consist of all the examinations and tests specified herein.

4.4.2 Production inspection.- Production units shall be inspected to the extent necessary to maintain quality control and conformance to the Government-accepted preproduction sample.

4.5 Examinations.- Examinations shall be performed to verify conformance to the following requirements:

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(a) Materials

(b) Construction

(c) Details of components

(d) Dimensions and tolerances

(e) Finish and color

(f) Nameplates or product markings

(g) Workmanship

(h) Preparation for delivery.

4.6 Tests.- The following test shall be performed:

(a) Operating test.

4.7 Test methods.-

4.7.1 Operating test.- The training aid shall be subjected to an operating test to insure proper functioning. During the operation period, not less than 12 calculations shall be performed to determine conformance to the requirements of 3.6.2.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C (see 6.2).

5.1.1 Level A.- For level A, each training aid shall be preserved and packaged for mechanical and physical protection in accordance with MIL-P-116, Method III.

5.1.1.1 Unit packaging.- Each training aid shall be packaged one unit to a container conforming to PPP-B-621.

5.1.2 Level C.- Each training aid shall be preserved and packaged to a degree which will afford protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity. This level, in general, will conform to applicable carrier rules and regulations. Suppliers commercial practices may be used when they meet the requirements of this level.

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5.2 Packing.- Packing shall be level A, B, or C (see 6.2).

5.2.1 Level A.- Units packaged in accordance with 5.1.1 shall be packed in exterior containers conforming to PPP-B-621, Class 2. Unless otherwise specified by the procuring activity (see 6.2), the gross weight shall not exceed the weight limitations of the shipping container. Each exterior shipping container shall be lined with a waterproof moisture barrier, sealed with adhesive or tape to effect water resistance in accordance with MIL-L-10547.

5.2.2 Level B.- Units packaged in accordance with 5.1.1 shall be packed in exterior containers conforming to PPP-B-621 and appendix thereto. Unless otherwise specified by the procuring activity (see 6.2), the gross weight shall not exceed the weight limitations of the shipping container.

5.2.3 Level C.- Units packaged as specified in 5.1.2 shall be packed in containers which will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with common carrier rules and regulations as applicable to the mode of transportation and may be the supplier's commercial practice when such meets the requirements of this level.

5.3 Physical protection.- All material shall be packaged and packed in containers constructed to provide a compact and nonshifting load. Adequate cushioning, blocking and bracing shall be provided within shipping containers in a manner designed to prevent damage due to shock, vibration or impact during shipment, handling and storage.

5.4 Marking.- Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The training aid will be used for classroom instruction in the use of the aerial navigation Dead Reckoning Computer, Type MB-4A.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification

(b) Procuring activity designation (device number or other) (see 1.1.1)

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(c) Selection of applicable levels of preservation, packaging and packing (see 5.1 and 5.2)

(d) Gross weight of shipping container (see 5.2.1)

(e) A Naval Training Device Center emblem will also be affixed to the training aid when specified by the procuring activity.

6.3 Definitions.- Except as specified herein, definitions will be furnsihed upon written request to the procuring activity.

6.3.1 Preproduction sample.- A production item furnished under production contracts for the purpose of determining compliance with requirements of the detail training device specification already established and accepted. It shall be electrically and mechanically representative of production sets though not necessarily made with production line jigs, fixtures and methods.

6.3.2 Production unit.- A unit in its final mechanical and electrical form of final production design made by production tools, jigs, fixtures and methods.

6.3.3 Training aid.- A training device designed to demonstrate or illustrate a concept, or to portray the functional characteristics of an end item without use of the actual working medium as a motivating force. Usually instructor operated and may also be used by the student for self-instruction.

Custodian:
Navy-TD
Air Force -70
Review Activity:
Air Force-70

Preparing activity:
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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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